THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · EASY LEVEL CHAPTER TEST SET 1

Chapter 10: **Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The ratio 12:18 in its simplest form is: 1

(a) 2:3 **(b)** 2:4 **(c)** 3:2 **(d)** 3:3

2. If 2:3 = 8:x, then x is: 1

(a) 7 **(b)** 16 **(c)** 12 **(d)** 11

3. ₹ 400 is divided between two friends in the ratio 3:5. The larger share (in ₹) is: 1

(a) 200 **(b)** 250 **(c)** 270 **(d)** 150

4. On a map, 1 cm represents 5 km. A distance of 7 cm on the map is actually (in km): 1

(a) 35 **(b)** 40 **(c)** 12 **(d)** 70

5. In a pie chart a sector represents 10% of the data. Its central angle (in degrees) is: 1

(a) 40 **(b)** 36 **(c)** 30 **(d)** 20

6. 4 workers can finish a job in 12 days. The number of days 8 workers (working at the same rate) need is: 1

(a) 8 **(b)** 20 **(c)** 6 **(d)** 48

7. The ratio 15:25 in its simplest form is: 1

(a) 5:3 **(b)** 3:6 **(c)** 3:5 **(d)** 4:5

8. If 3:5 = 18:x, then x is: 1

(a) 23 **(b)** 30 **(c)** 36 **(d)** 11

9. ₹ 360 is divided between two friends in the ratio 4:5. The larger share (in ₹) is: 1

(a) 220 **(b)** 180 **(c)** 160 **(d)** 200

10. On a map, 1 cm represents 10 km. A distance of 6 cm on the map is actually (in km): 1

(a) 70 **(b)** 16 **(c)** 60 **(d)** 54

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. If 6 pens cost ₹ 90, find the cost of 15 pens. 2

12. 10 men can build a wall in 24 days. In how many days can 8 men build it? 2

13. Divide ₹ 840 between A and B in the ratio 3:4. 2

14. If 8 pens cost ₹ 120, find the cost of 20 pens. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The scale of a map is 1 cm = 12 km. Two cities are 5 cm apart on the map. Find the actual distance. Also find the map distance of two towns that are actually 96 km apart. 3

16. Divide ₹ 1200 among A, B and C in the ratio 2:3:5. 3

17. In a pie chart of 200 students' favourite games, cricket takes up 30% of the chart. Find the central angle of the cricket sector and the number of students who like cricket. 3

OR

The scale of a map is 1 cm = 30 km. Two cities are 4 cm apart on the map. Find the actual distance. Also find the map distance of two towns that are actually 150 km apart. 3

18. Divide ₹ 900 among A, B and C in the ratio 1:2:3. 3

OR

In a pie chart of 240 students' favourite games, cricket takes up 25% of the chart. Find the central angle of the cricket sector and the number of students who like cricket. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The data of 30 students' choices is: Cricket 12, Football 9, Tennis 6, Hockey 3. Find the central angle of each sector for a pie chart and verify that the angles add to 360°. 5

OR

A camp has food for 20 people for 15 days. If 10 more people join the camp, for how many days will the same food last? 5

20. The data of 45 students' choices is: Maths 15, Science 12, English 9, Hindi 9. Find the central angle of each sector for a pie chart and verify that the angles add to 360°. 5

OR

A camp has food for 30 people for 12 days. If 6 more people join the camp, for how many days will the same food last? 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND